Processor/Bricklet

U101A
XMC1404
U101B
XMC1404
U101C
XMC1404
U101D
XMC1404
U101E
XMC1404
SW101
SW102
OS103011MS8QP1
Boot
J101
blue
D103
"status led"
1k
R124
C12816MHz
10pF
C129
X101
10pF
P112
DC-Protect
RP103
R1
C107
1uF
FB
C103
L101
C106
1uF
1uF
U110
TPS70933DBVR
Ersatz TPS70933QDBVRQ1
C110
2.2uF
C111
100uF
5V->3.3V
Ideal Diode 5V Power for ESP
Q105
SSM3J355R,LF
U103
DMMT5401
FB
L103
P101
CON-SENSOR2
RP101
82C101
220pF
10k
R140
47k
R153
Maximum Power Setting
no jumper: invalid
P0_0 = 3V3, P0_5 = 3V3 -> invalid
P0_0 = DNC, P0_5 = 3V3 -> 6A
P0_0 = GND, P0_5 = 3V3 -> 10A
P0_0 = 3V3, P0_5 = DNC -> 13A
P0_0 = DNC, P0_5 = DNC -> 16A
P0_0 = GND, P0_5 = DNC -> 20A
P0_0 = 3V3, P0_5 = GND -> 25A
P0_0 = DNC, P0_5 = GND -> 32A
P0_0 = GND, P0_5 = GND -> reserved
SPI Slave/CH0
P0.13 : USIC0_CH0-DX2F : SEL
P0.14 : USIC0_CH0-DX1A : CLK
P0.15 : USIC0_CH0-DX0B : MOSI
P2.0 : USIC0_CH0-DOUT0 : MISO

AC/DC

P104
Power
P106
L'+PE
F102
FUSE
D115
1N4007
D116
1N4007
R118
RNF12FTD249K
R116
RNF12FTD249K
U109
HCPL2731
RP105
10k
P103
L'+PE
Phoenix
1755736
1757019
F101
FUSE
R128
MOV-10D391KTR
U111
IRM02-12
C105
4.7uF/50V
C124
1uF
C125
1uF
U112
WPMDL110
C126
1uF

RS485

RP102
10k
U113
SP3491
C132
100nF
750
R134
DNP
R135
750
R136
SMAJ24CA
SMAJ24CA
D111
L104A
L104B
L104C
L104D
WE-SL
R137
120
PTC 0.45A
PTC101
0ZC.0016FF2E
P110
MODBUS

CP/PP

L102
NR3015T6R8M
C104
4.7uF/50V
U102
PEM1-S12-D12-S
C108
4.7uF/50V
C109
4.7uF/50V
-12V
+12V
R129
300k
R130
100k
R102
75k
R104
300k
R122
100k
Q103
2N7002P
R138
1k
D112
B0520LW-7
REL101
AGQ200S12Z
Vcpadc = (75*Vcp+990)/600
Vcpadc = Vcp-300k/100k*Vcpadc+300k/75k*3.3V-300k/75k*Vcpadc
Option
0402 1NF 100V COG
R125
75k
R126
300k
R127
100k
C120
DNP
R110
75k
R108
300k
R109
100k
C121
DNP
R105
100k
R106
100k
R103
10k
R101
100
OPA171AIDBVR
U105
C112
100nF/30V
C113
100nF/30V
R107
910
öffner
SMF4L13CA
D105
P105
CP/PP/PE
R111
1k
R112
10k
R141
2k
D107
PESD3V3L1BA
Rpp = (VPP*1k*2k)/(5V*2k-VPP*(1k+2k))

12V Input/Motor/GP

C116
100nF/30V
C117
100nF/30V
C118
100nF/30V
C119
100nF/30V
C114
10uF
12V
U106
A5950GLP
RP104
1k
P107
Motor
SMF4L13CA
D108
SMF4L13CA
D109
R119
10k
R117
1k
D104
P109
Enable
R131
1k
R132
10k
D102
P113
GP
R123
1k
R139
10k
Q104
2N7002P
D113
D114
P108
Switch_LED
Q101
2N7002P
D101
R120
1k
R121
10k
D110

Relay

P102
135° 1884340000
180° 1825960000
MOV-10D391KTR
R113
AQH3213AX
U108
R114
120
Q102
SI2305CDS-T1-GE3
R115
1k

Rail DRILL DRILL
Rail DRILL DRILL

Copyright (©) 2021, B.Nordmeyer <bastian@tinkerforge.com>
Licensed under CERN OHL v.1.1

Tinkerforge GmbH

Sheet: /
File: working.kicad_sch

Title: EVSE Bricklet

Size: A3	Date: 2023-01-20	Rev: 2.0.6
KiCad E.D.A. eeschema 7.0.6		Id: 1/1

Copyright Tinkerforge GmbH 2021.
This documentation describes Open Hardware and is licensed under the CERN OHL v. 1.1.
You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (http://ohwr.org/cernohl). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions